Computer Networks

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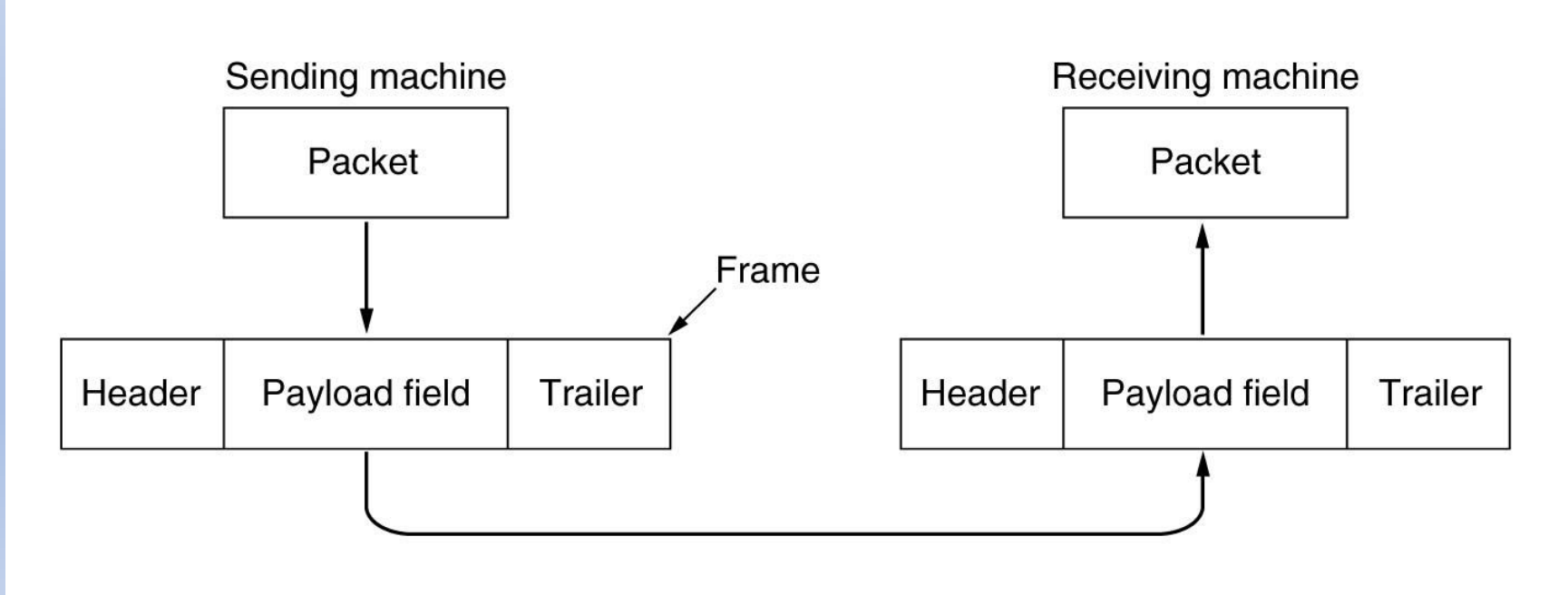
November 26, 2024 (Tuesday)

Lecture # 27

Link Layer

Reliable Transmission Over Unreliable Physical Link



Error Control - detection and correction of errors

- Two types of errors are:
  - lost frames : fails to arrive at receiver
  - damaged frames : recognizable frame arrives, but bits damaged

- Common techniques for error control are based on :
  - error detection
  - **positive acknowledgment:** returned by destination if successfully received error free frames
  - **retransmission** after timeout
  - **negative acknowledgement** & retransmission

Automatic Repeat Request (ARQ)

Collective name for such error control mechanisms.

Most Error Control techniques are based on

(1) Error Detection Scheme (e.g., Parity checks, CRC), and

(2) Retransmission Scheme

Error control schemes that involve error detection and retransmission of lost or corrupted frames are referred to as ARQ error control.

Automatic Repeat Request (ARQ)

1. Stop-and-Wait ARQ
2. Go-Back-N ARQ
3. Selective Repeat ARQ

The protocol for sending ACKs in all ARQ protocols are based on the sliding window flow control scheme.

Stop and Wait ARQ

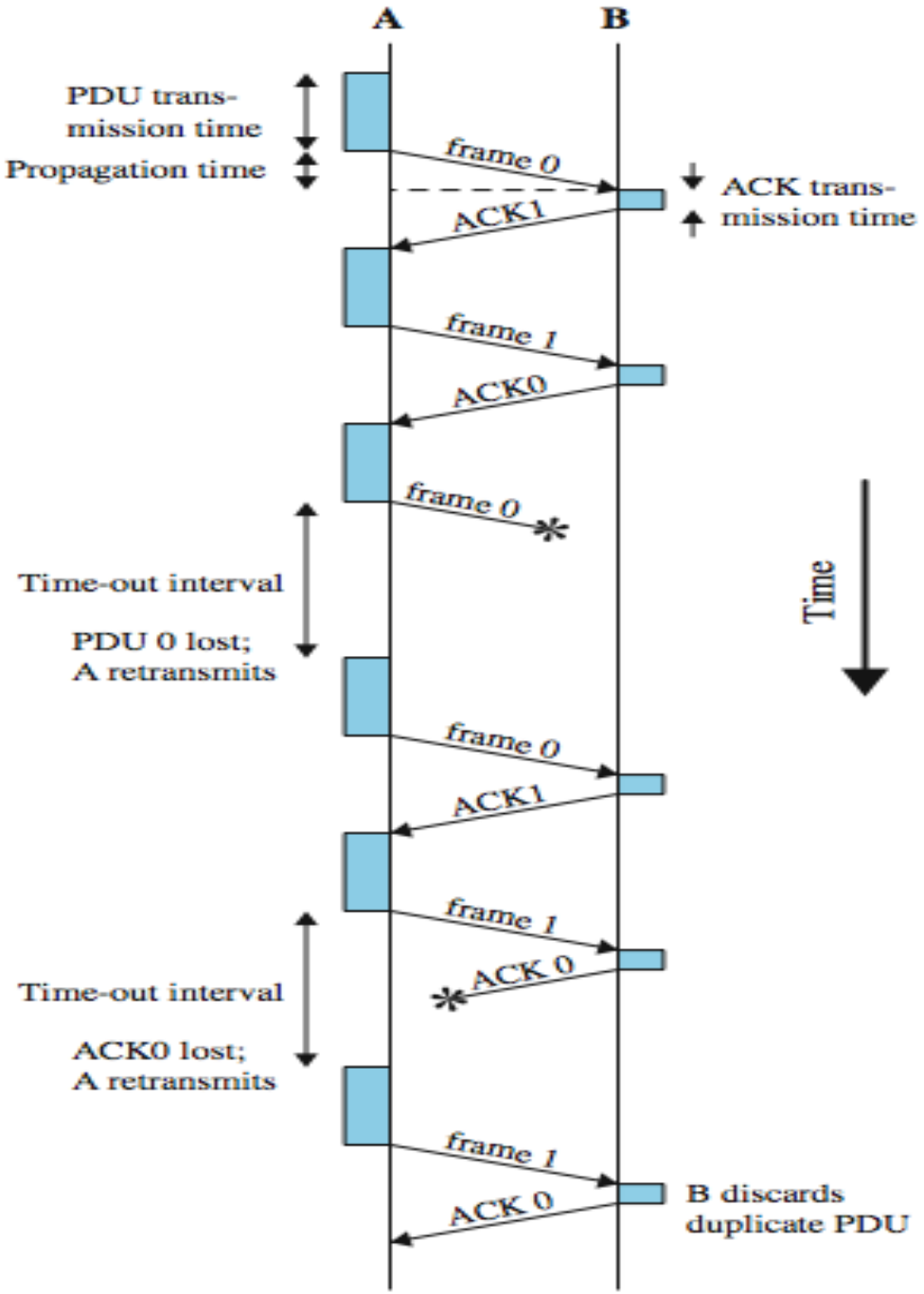
- Source transmits single frame
- Wait for ACK
- if received frame damaged, discard it & no ACK
  - transmitter waits till timeout
  - if no ACK within timeout, retransmit
- if ACK damaged, transmitter will not recognize it
  - transmitter will retransmit
  - receiver gets two copies of frame

Stop and Wait ARQ

To avoid this problem, frames are alternatively labeled with 0 or 1.

i.e., Frames have 1-bit sequence numbers (SN = 0 or 1)

- positive acknowledgment are ACK0 / ACK1
- ACK0 – frame numbered 1……



Go-Back-N ARQ (continuous ARQ)

- Based on sliding window
- If no error, the operations are identical to Sliding Window. Use window to control number of outstanding frames

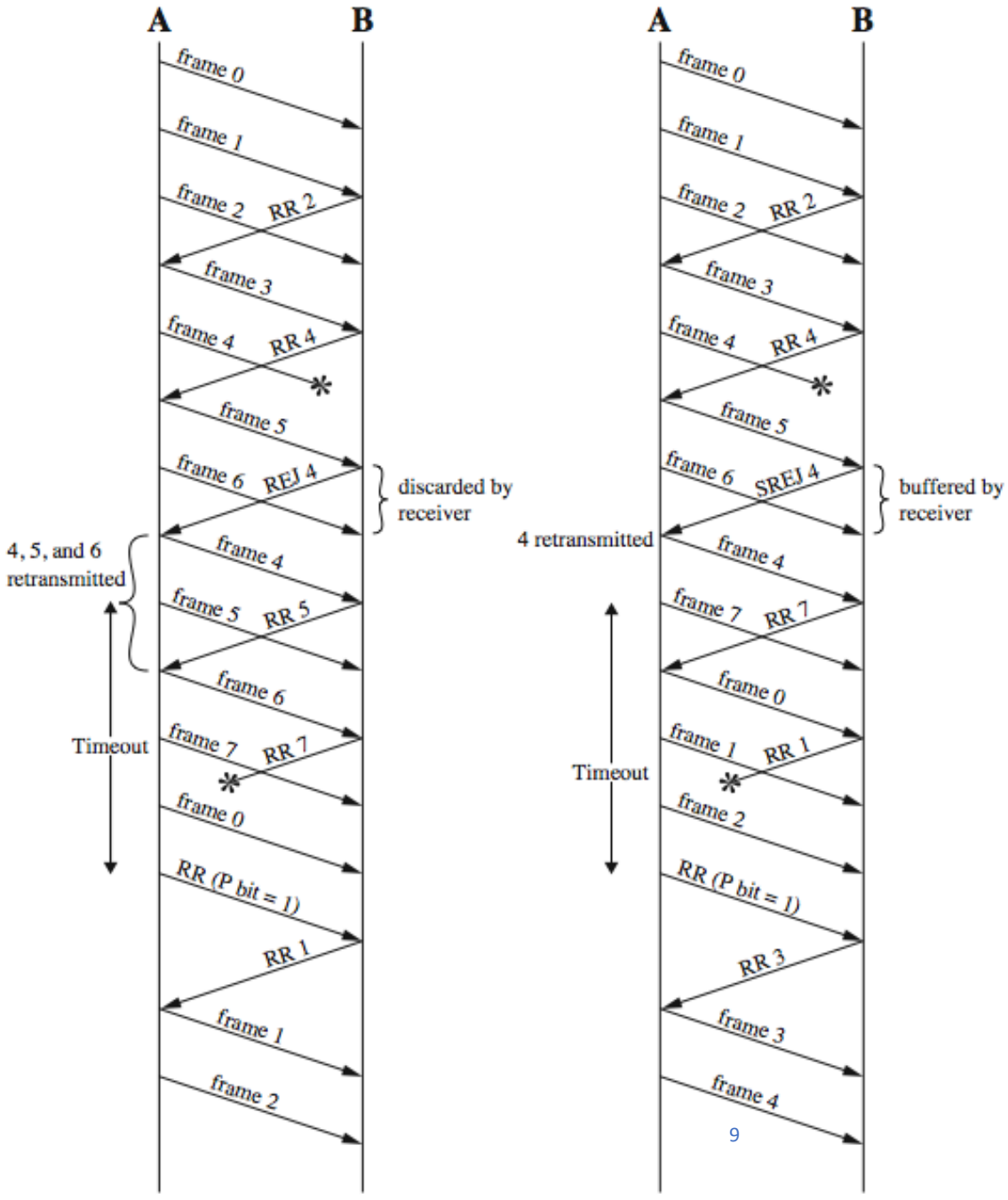
A station may send multiple frames as allowed by the window size

a. Receiver sends a REJ i if frame i is in error. After that, the receiver discards all incoming frames until the frame in error was correctly retransmitted
b. If sender receives a REJ i it will retransmit frame i and all packets i+1, i+2,... which have been sent, but not been acknowledged

Selective Reject ARQ

- Also called selective retransmission
- Only rejected frames are retransmitted
- Subsequent frames are accepted by the receiver and buffered
- Minimizes retransmission
- Receiver must maintain large enough buffer
- More complex logic in transmitter, hence less widely used
- useful for satellite links with long propagation delays

Go Back N versus Selective Reject



(a) Go-back-N ARQ

(b) Selective-reject ARQ

Link layer, LANs

- introduction
- error detection, correction
- multiple access protocols
- LANs
  - addressing, ARP
  - Ethernet
  - switches

- a day in the life of a web request

Multiple access links, protocols

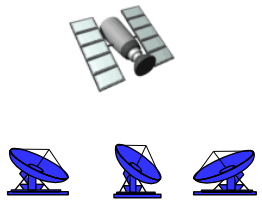
two types of "links":

- point-to-point
  - point-to-point link between Ethernet switch, host
  - PPP for dial-up access
- broadcast (shared wire or medium)
  - old-fashioned Ethernet
  - upstream HFC in cable-based access network
  - 802.11 wireless LAN, 4G/4G. satellite

shared wire (e.g., cabled Ethernet)

shared radio: 4G/5G

shared radio: WiFi



shared radio: satellite

Multiple access protocols

- single shared broadcast channel
- two or more simultaneous transmissions by nodes: interference
  - *collision* if node receives two or more signals at the same time

multiple access protocol

- distributed algorithm that determines how nodes share channel, i.e., determine when node can transmit
- communication about channel sharing must use channel itself!
  - no out-of-band channel for coordination

An ideal multiple access protocol

given: multiple access channel (MAC) of rate *R* bps

desiderata:

1. when one node wants to transmit, it can send at rate *R*.
2. when M nodes want to transmit, each can send at average rate *R/M*
3. fully decentralized:
   - no special node to coordinate transmissions
   - no synchronization of clocks, slots
4. simple

MAC protocols: taxonomy

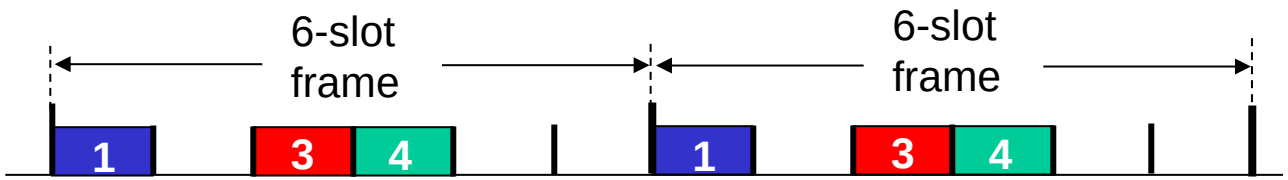
three broad classes:

- channel partitioning
  - divide channel into smaller “pieces” (time slots, frequency, code)
  - allocate piece to node for exclusive use
- *random access*
  - channel not divided, allow collisions
  - “recover” from collisions
- “taking turns”
  - nodes take turns, but nodes with more to send can take longer turns

Channel partitioning MAC protocols: TDMA

TDMA: time division multiple access

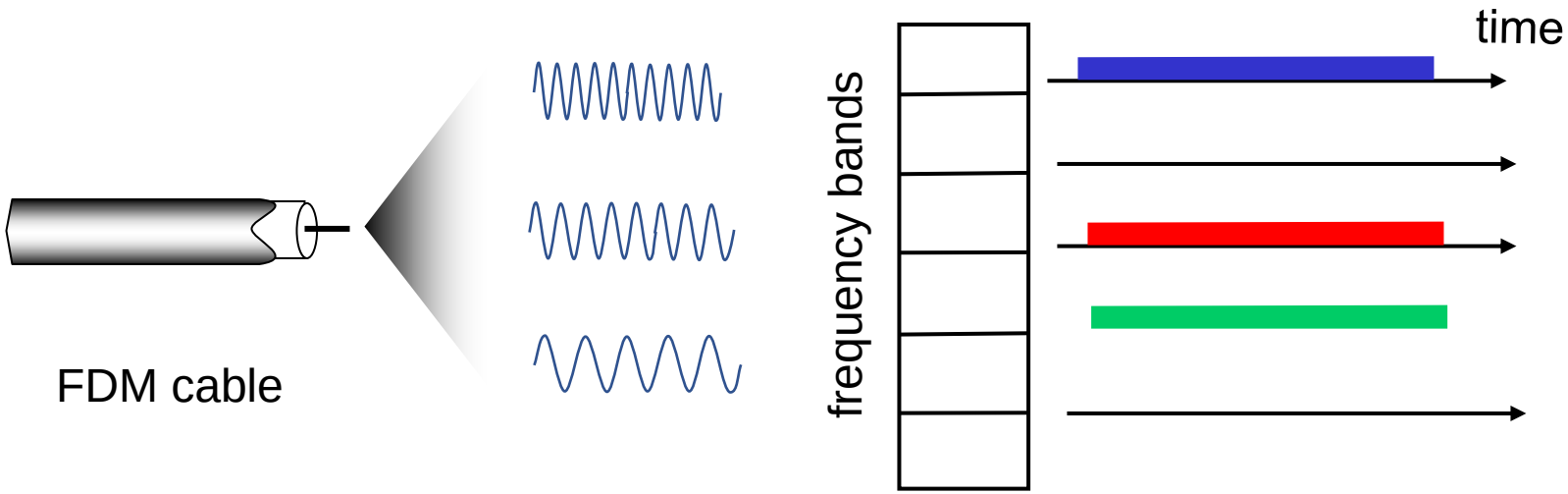
- access to channel in “rounds”
- each station gets fixed length slot (length = packet transmission time) in each round
- unused slots go idle
- example: 6-station LAN, 1,3,4 have packets to send, slots 2,5,6 idle



Channel partitioning MAC protocols: FDMA

FDMA: frequency division multiple access

- channel spectrum divided into frequency bands
- each station assigned fixed frequency band
- unused transmission time in frequency bands go idle
- example: 6-station LAN, 1,3,4 have packet to send, frequency bands 2,5,6 idle



Random access protocols

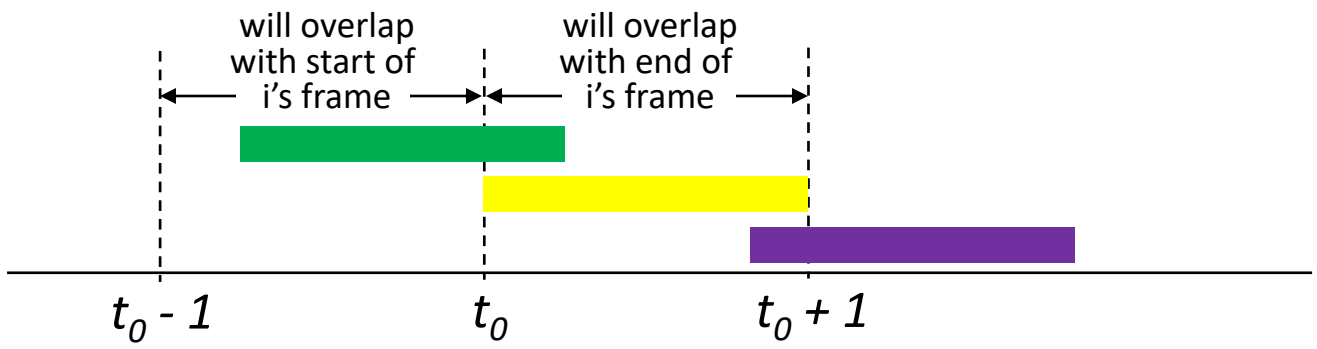
- when node has packet to send
  - transmit at full channel data rate R.
  - no *a priori* coordination among nodes
- two or more transmitting nodes: “collision”
- random access MAC protocol specifies:
  - how to detect collisions
  - how to recover from collisions (e.g., via delayed retransmissions)
- examples of random access MAC protocols:
  - ALOHA, slotted ALOHA
  - CSMA, CSMA/CD, CSMA/CA

Pure ALOHA

- A station transmits the data whenever there is data to be sent.
- Then, the station listens to the channel to see if a collision occurred.
- If the frame was destroyed, the station waits for a random length of time and tries again.
- Systems in which multiple users share a common channel in a way that can lead to conflicts are widely known as **contention** systems.

Pure ALOHA

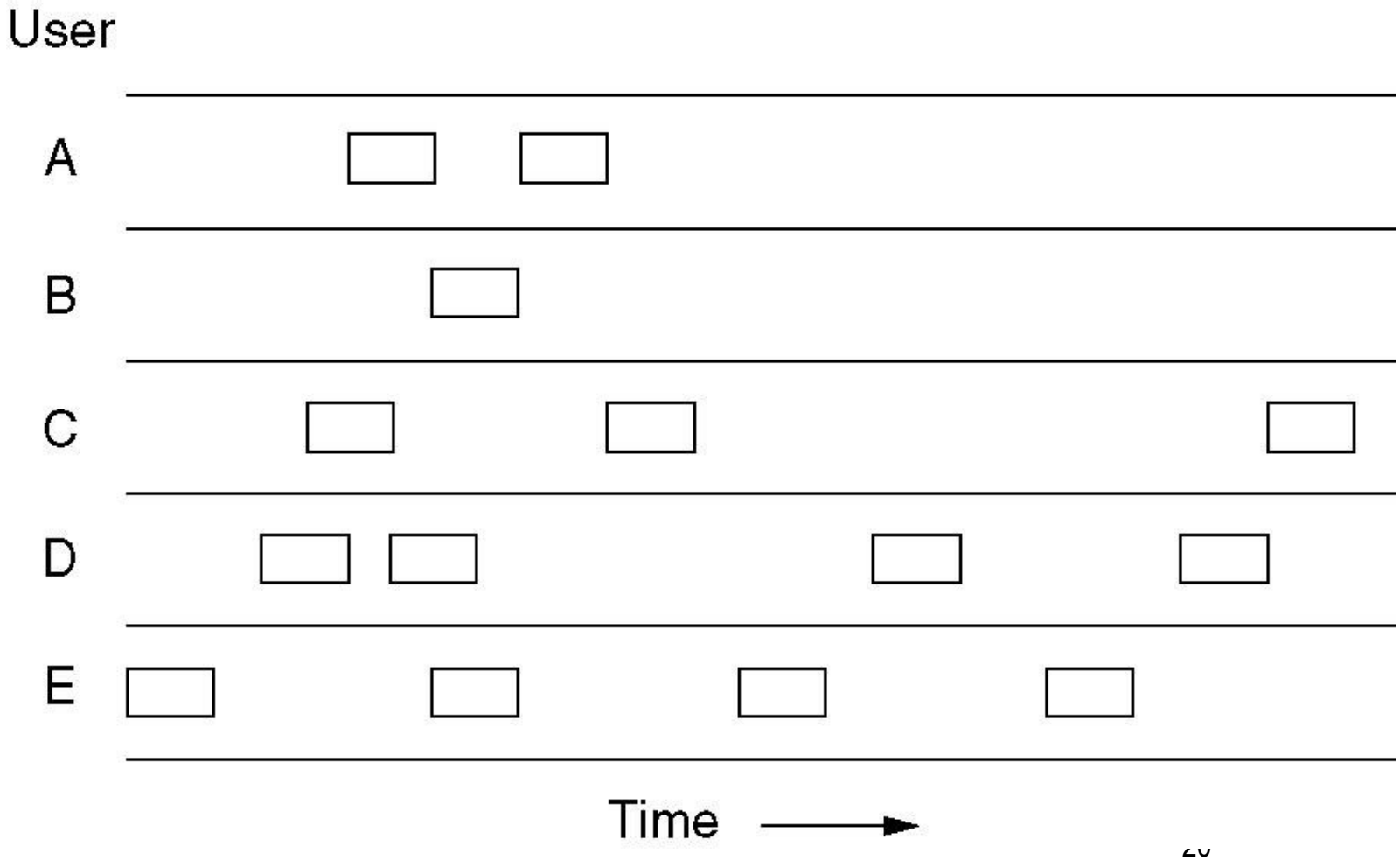
- unslotted Aloha: simpler, no synchronization
  - when frame first arrives: transmit immediately
- collision probability increases with no synchronization:
  - frame sent at t_0 collides with other frames sent in $[t_0-1, t_0+1]$



- pure Aloha efficiency: 18% !

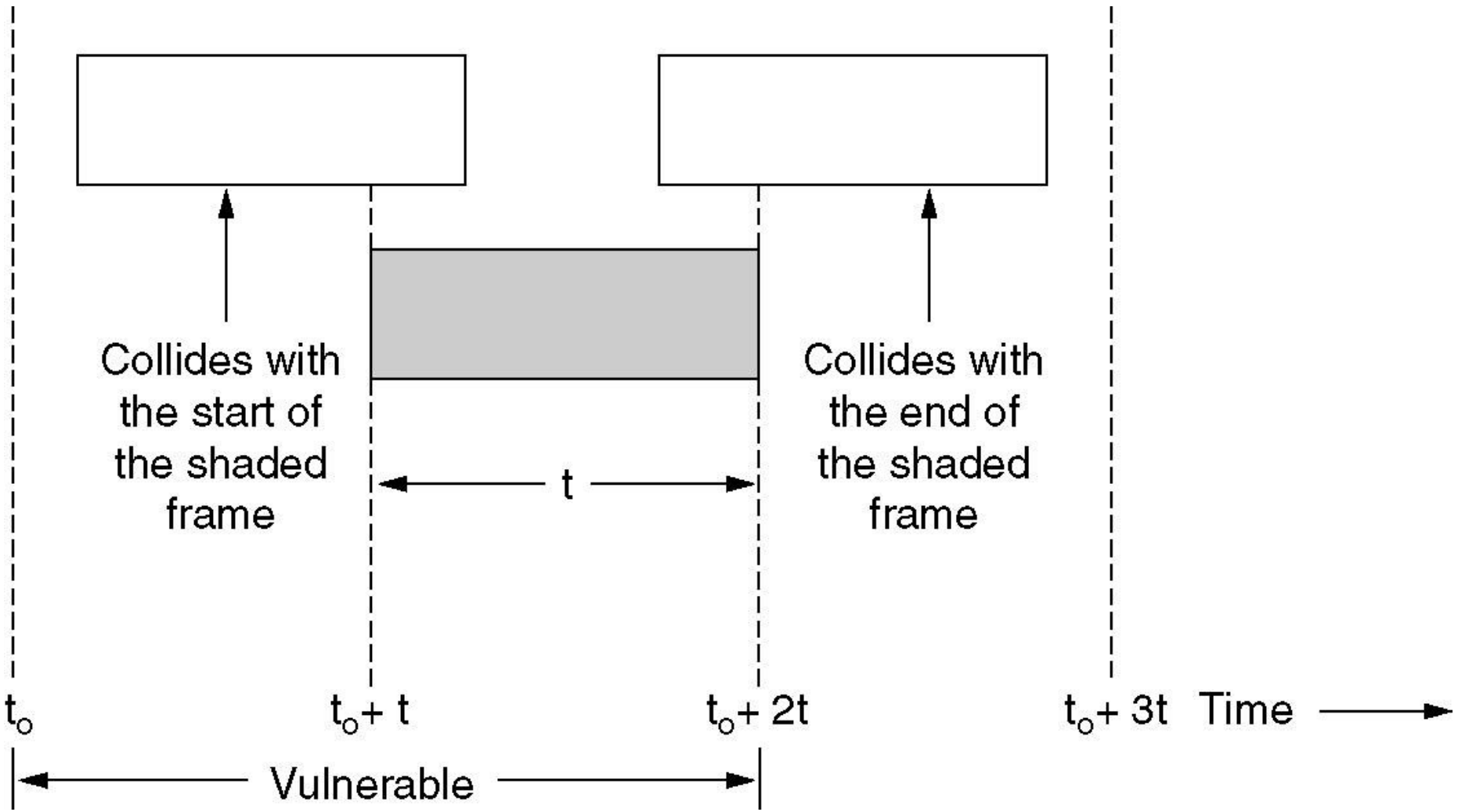
Pure ALOHA

In pure ALOHA, frames are transmitted at completely arbitrary times.



Pure ALOHA

Vulnerable period for the shaded frame.



Slotted ALOHA

- A computer is not permitted to send whenever a carriage return is typed but wait for a time slot.

- Time is divided into fixed slots of one frame time each. A station waits until the start of the next slot before transmitting a frame. Thus, $P_0 = e^{-G}$ (the vulnerable period is only one time slot). $S = N e^{-G}$ Note that as N increases, the number of collisions increases exponentially.

- Note, the maximum occurs when N = 1. S = 0.368.
- Slotted Aloha can be used to allocate a shared cable channel.

Slotted ALOHA

assumptions:

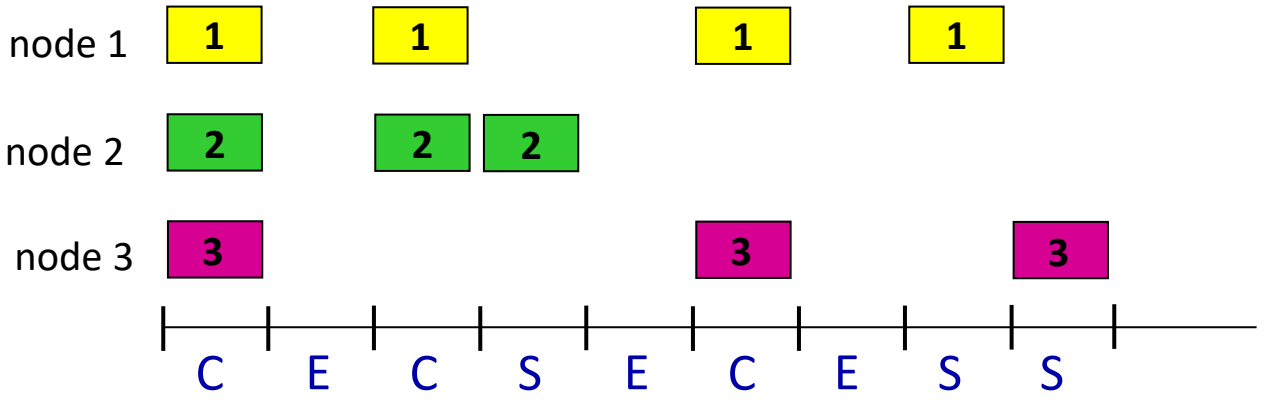
- all frames same size
- time divided into equal size slots (time to transmit 1 frame)
- nodes start to transmit only slot beginning
- nodes are synchronized
- if 2 or more nodes transmit in slot, all nodes detect collision

operation:

- when node obtains fresh frame, transmits in next slot
  - *if no collision:* node can send new frame in next slot
  - *if collision:* node retransmits frame in each subsequent slot with probability *p* until success

randomization – *why*?

Slotted ALOHA



C: collision
S: success
E: empty

Pros:

- single active node can continuously transmit at full rate of channel
- highly decentralized: only slots in nodes need to be in sync
- simple

Cons:

- collisions, wasting slots
- idle slots
- nodes may be able to detect collision in less than time to transmit packet
- clock synchronization

Slotted ALOHA: efficiency

efficiency: long-run fraction of successful slots (many nodes, all with many frames to send)

- *suppose: N* nodes with many frames to send, each transmits in slot with probability *p*
  - prob that given node has success in a slot = $p(1-p)^{N-1}$
  - prob that *any* node has a success = $Np(1-p)^{N-1}$
  - max efficiency: find p^* that maximizes $Np(1-p)^{N-1}$
  - for many nodes, take limit of $Np^*(1-p^*)^{N-1}$ as *N* goes to infinity, gives:

  max efficiency = 1/e = .37

- *at best:* channel used for useful transmissions 37% of time!

CSMA (carrier sense multiple access)

simple CSMA: listen before transmit:

- if channel sensed idle: transmit entire frame
- if channel sensed busy: defer transmission

- human analogy: don't interrupt others!

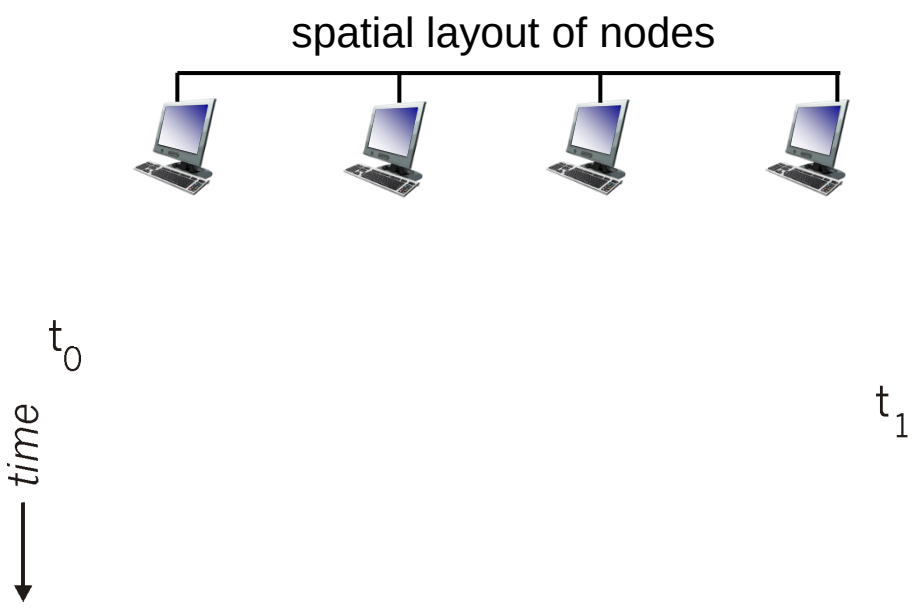
CSMA/CD: CSMA with *collision detection*

- collisions *detected* within short time
- colliding transmissions aborted, reducing channel wastage
- collision detection easy in wired, difficult with wireless

- human analogy: the polite conversationalist

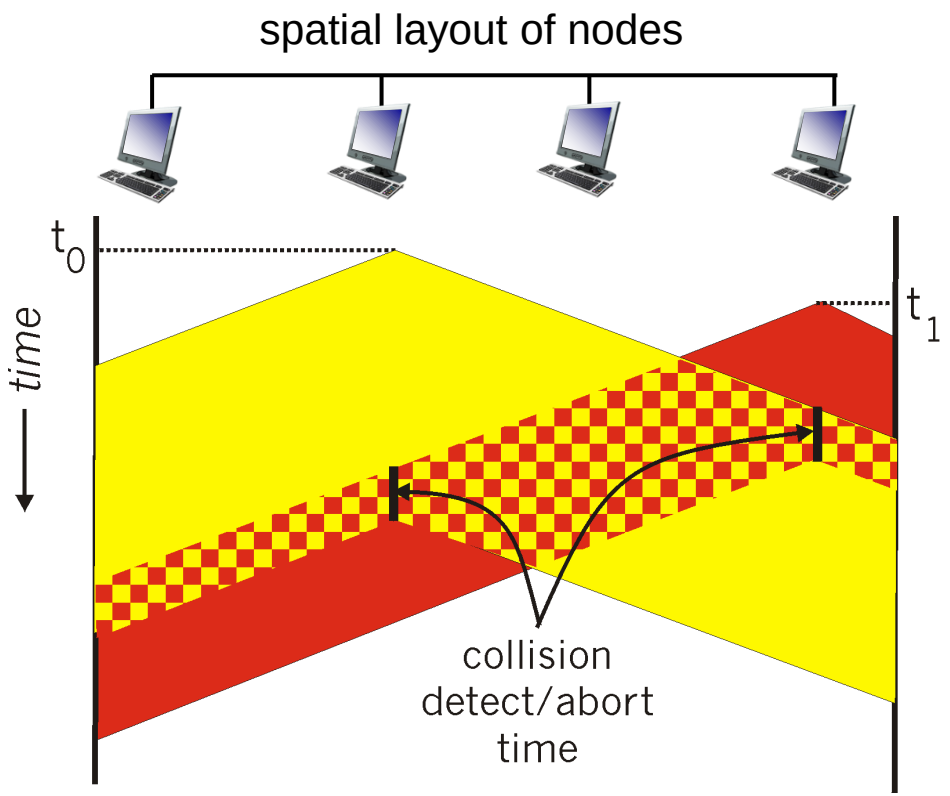
CSMA: collisions

- collisions *can* still occur with carrier sensing:
  - propagation delay means two nodes may not hear each other's just-started transmission
- collision: entire packet transmission time wasted
  - distance & propagation delay play role in in determining collision probability



CSMA/CD:

- CSMA/CS reduces the amount of time wasted in collisions
  - transmission aborted on collision detection



Ethernet CSMA/CD algorithm

1. NIC receives datagram from network layer, creates frame
2. If NIC senses channel:

   if idle: start frame transmission.

   if busy: wait until channel idle, then transmit
3. If NIC transmits entire frame without collision, NIC is done with frame !
4. If NIC detects another transmission while sending: abort, send jam signal
5. After aborting, NIC enters *binary (exponential) backoff:*
   - after *m*th collision, NIC chooses *K* at random from $\{0,1,2, \ldots, 2^m-1\}$. NIC waits $K \cdot 512$ bit times, returns to Step 2
   - more collisions: longer backoff interval

The Binary Exponential Backoff Algorithm

- If collision
  - Time is divided into discrete slots whose length is equal to the worst-case round-trip propagation time on the ether (2τ).
    - minimum frame is 64 bytes (header + 46 bytes of data) = 512 bits
    - Channel capacity 10 Mbps, 512/10 M = 51.2μ
  - delay and try again (**binary exponential backoff**)
    - 1st time: waits 0 or 1 slotted time (51.2μs)
    - 2nd time: waits 0, 1, 2, or 3 slotted time
    - nth time: k slotted time, for randomly selected $k=0..2^n - 1$
    - Frozen at 1023 slots and give up after several tries (usually 16)

CSMA/CD efficiency

- T_{prop} = max prop delay between 2 nodes in LAN
- t_{trans} = time to transmit max-size frame

$$efficiency = \frac{1}{1 + 5t_{prop}/t_{trans}}$$

- efficiency goes to 1
  - as t_{prop} goes to 0
  - as t_{trans} goes to infinity
- better performance than ALOHA: and simple, cheap, decentralized!

“Taking turns” MAC protocols

channel partitioning MAC protocols:

- share channel *efficiently* and *fairly* at high load
- inefficient at low load: delay in channel access, 1/N bandwidth allocated even if only 1 active node!

random access MAC protocols

- efficient at low load: single node can fully utilize channel
- high load: collision overhead

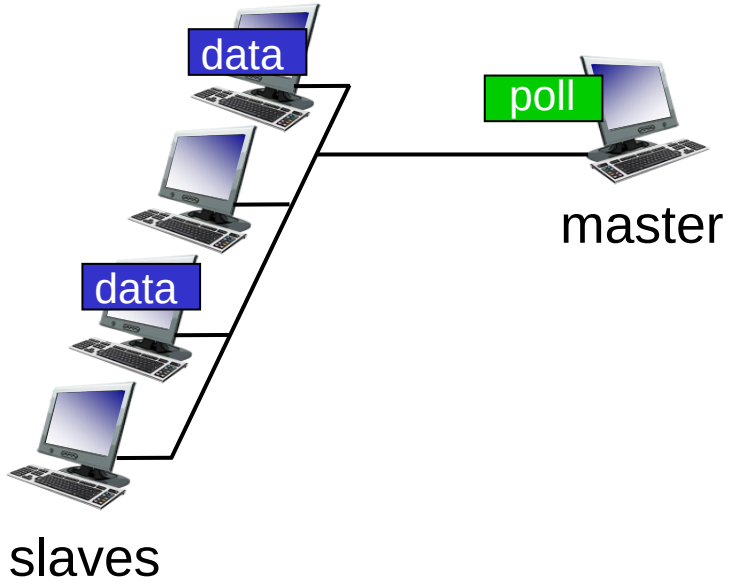
“taking turns” protocols

- look for best of both worlds!

“Taking turns” MAC protocols

polling:

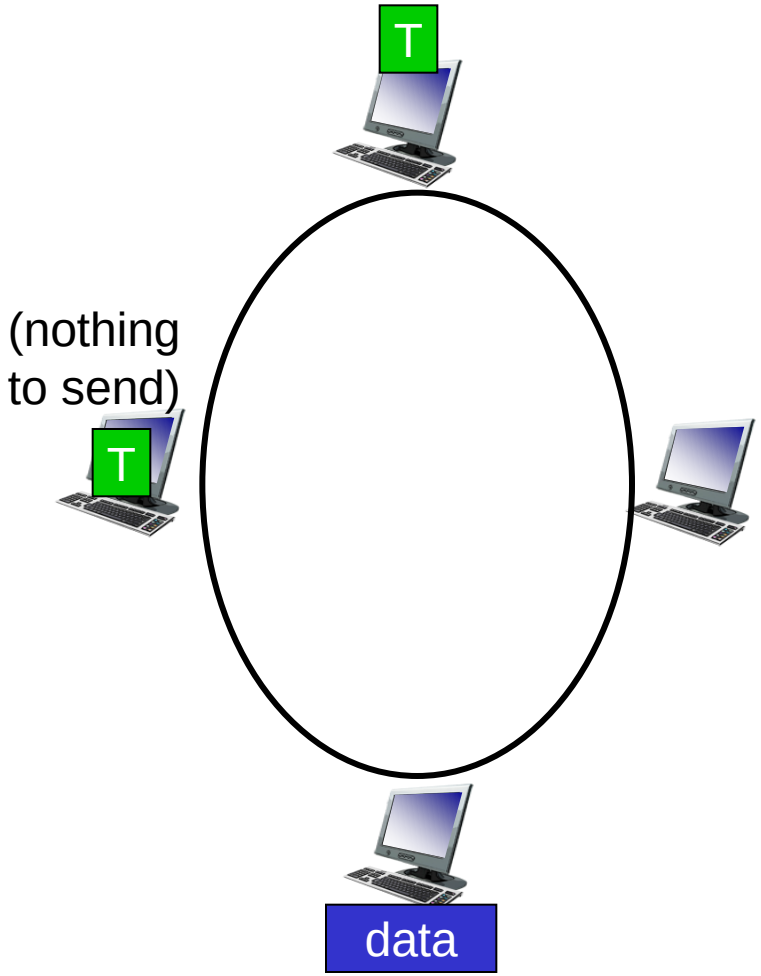
- master node “invites” other nodes to transmit in turn
- typically used with “dumb” devices
- concerns:
  - polling overhead
  - latency
  - single point of failure (master)



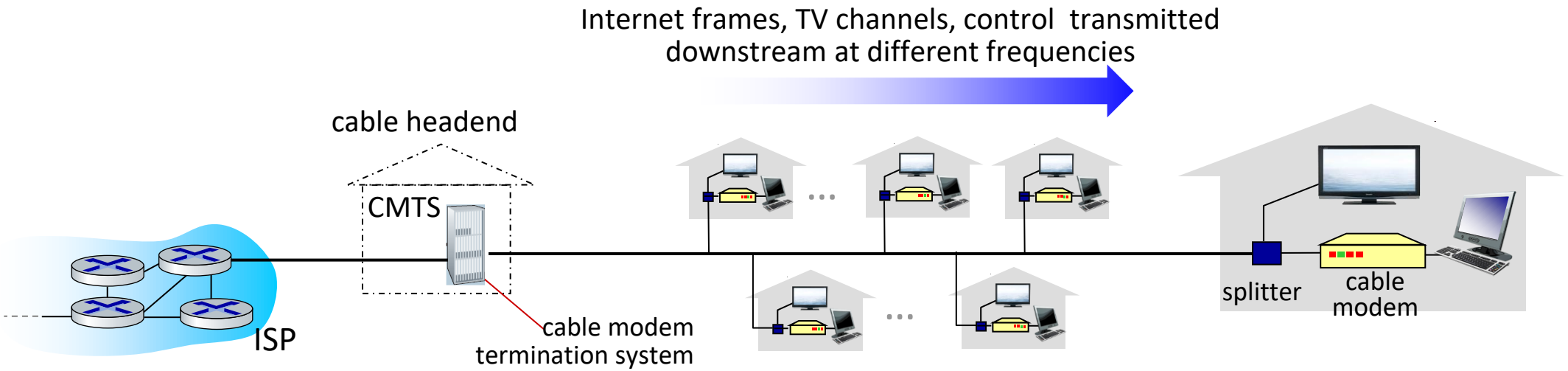
“Taking turns” MAC protocols

token passing:

- control *token* passed from one node to next sequentially.
- token message
- concerns:
  - token overhead
  - latency
  - single point of failure (token)



Cable access network: FDM, TDM *and* random access!



- multiple downstream (broadcast) FDM channels: up to 1.6 Gbps/channel
  - single CMTS transmits into channels
- multiple upstream channels (up to 1 Gbps/channel)
  - multiple access: all users contend (random access) for certain upstream channel time slots; others assigned TDM

Summary of MAC protocols

- channel partitioning, by time, frequency or code
  - Time Division, Frequency Division
- random access (dynamic),
  - ALOHA, S-ALOHA, CSMA, CSMA/CD
  - carrier sensing: easy in some technologies (wire), hard in others (wireless)
  - CSMA/CD used in Ethernet
  - CSMA/CA used in 802.11
- taking turns
  - polling from central site, token passing
  - Bluetooth, FDDI, token ring